Behavioural Analytics Hackathon

Problem Statement 1:
Early Detection of Student Burnout & Dropout Risk
Background
Universities often identify academic issues only after performance drops. By that time, intervention becomes difficult. Early behavioural signals such as reduced LMS activity, delayed submissions, and attendance changes can indicate burnout risk.
Challenge
Build a behavioural analytics system that predicts:

- Burnout Risk Level (Low / Medium / High)
- Academic Disengagement Indicators
- Dropout Probability

Possible Data Inputs
Teams may use or simulate:

- LMS login frequency
- Assignment submission delays
- Attendance trends
- Sentiment analysis from feedback forms
- Activity pattern irregularities

Expected Output
Your solution must provide:

- A Risk Score (0–100 or categorical classification)
- Key behavioural triggers influencing the prediction
- Recommended intervention strategy
- Visualization dashboard (optional but encouraged)

Problem Statement 2
Detecting Financial Impulse Behaviour in Young Adults
Background
Young individuals often overspend due to emotional or impulsive decision-making. Identifying behavioural spending patterns can help predict risky financial behaviour.
Challenge
Design a system that:
- Detects impulsive spending patterns
- Identifies emotional or timing-based triggers
- Predicts high-risk upcoming spending behaviour

Possible Input Features
- Transaction timestamps
- Merchant categories
- Spending frequency spikes
- End-of-month spending surge
- Behavioural clustering patterns

Expected Output
Your system should generate:
- Impulse Risk Score
- Spending Behaviour Profile
- Personalized behavioural nudges or recommendations

Problem Statement 3
Fake Engagement Detection in Social Media
Background
Social media platforms often experience fake engagement generated by bots or coordinated activity. This distorts genuine behavioural patterns.
Challenge
Build a behavioural detection model that:
- Differentiates organic versus artificial engagement
- Detects coordinated behavioural anomalies

Behavioural Indicators
- Timing regularity
- Engagement burst patterns
- Network interaction patterns
- Linguistic consistency

Expected Output
- Authenticity Score
- Bot Probability
- Behavioural anomaly explanation

Problem Statement 4

Open Track (Behavioural Analytics Focus):

- Those who are opting for the Open Track must build a solution strictly related to Behavioural Analytics.

- The solution must involve behavioural pattern analysis, predictive modelling, or behavioural decision intelligence.

- Possible domains include (but are not limited to):
  - Human decision-making modelling
  - Behaviour-driven risk prediction systems
  - Consumer behaviour analytics
  - Digital behaviour anomaly detection.

1. Event Timeline:

Start Date & Time: Saturday, 28 February – 9:00 AM
End Date & Time: Sunday, 1 March – 11:59 PM
Note: Submissions after 11:59 PM will not be considered.

2. Submission Requirements (Mandatory)

A. GitHub Repository

The repository must include:
- Clean and well-structured source code
- Dataset or clearly documented simulation process
- README file
- **If using synthetic data:**

Mention:

- Dataset type: Synthetic
- Why (no real behavioural dataset available)
- How you generated it (rules, distributions, assumptions)
- Number of records
- Feature description

If using public data:

- Dataset source + link
- Why it fits behavioural analytics
- What behavioural features you engineered from it

The GitHub repository must be public and accessible.

B. Model Explanation Document (Maximum 5 Pages)

The document must include:
- Problem understanding
- Data assumptions / dataset description
- Feature engineering logic
- Model selection and reasoning
- Evaluation metrics
- Behavioural insights derived

C. Final Presentation (PPT)

A presentation template will be shared separately.

4. Evaluation Criteria

- Model performance and validation
- Feature engineering innovation
- Depth of behavioural insight
- Practical feasibility
- Clarity of explanation
- Visualization and presentation quality

5. Important Rules

- Data may be simulated if real data is not available.
- Any ML/AI approach is allowed.
- Clearly explain all assumptions.
- Plagiarism or copied repositories will lead to disqualification.
- Late submissions will not be evaluated.

6. Final Submission Format

Submit GitHub Repository link as open access repository.
The submission details of Github Repository will be shared by FEB 28 – 3 PM.